

Loan Approval Prediction Using Machine Learning

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Abstract

This report presents a two-step machine learning approach to assess credit risk through Probability of Default (PD) Modeling and Expected Loss (EL) Estimation. The dataset consists of historical loan applications with various applicant attributes, such as income, credit score and employment status. Several preprocessing steps, including feature scaling and encoding, were performed to optimize model performance. In Step 1, multiple models including Logistic Regression, LightGBM, Random Forest and XGBoost were tested to predict the probability of default. Step 2 builds on the predicted PD values to estimate expected financial losses using Lasso Regression, XGBoost and Neural Networks, with and without Principal Component Analysis (PCA). The results highlight the robustness of XGBoost in handling structured financial datasets, outperforming traditional statistical models. Future work could explore hybrid models integrating deep learning techniques, enhanced feature engineering, as well as real-time adaptation to further refine credit risk assessment.

1 Introduction

The global lending industry is valued at over \$8 trillion, with banks and financial institutions processing millions of loan applications annually (Statista, 2023). However, manual loan approval processes remain inefficient, biased and time-consuming. Traditional underwriting relies mainly on credit scores, debt-to-income ratios as well as employment history. This may not always reflect the applicant's true repayment ability.

1.1 Current Loan Approval Challenges

Bias in Loan Approval: Minority applicants face higher rejection rates than their white counterparts, even when they possess similar financial profiles. Research by Harvard Business Review (2021) highlights that minority applicants were 40% more likely to be denied loans. This bias, whether intentional or systemic, limits economic opportunities for underserved communities. Leveraging machine learning for credit assessments can reduce subjective decision-making and ensure fairness in loan approvals.

High Loan Default Rates: Loan defaults present significant risks to financial institutions. The global average loan default rate ranges between 2-5%, but certain segments, such as small business loans, see default rates exceeding 10% (Federal Reserve, 2023). AI-driven credit risk models offer a predictive edge, potentially reducing bad loan rates by 20% and bolstering lender profitability (McKinsey & Co., 2022).

Slow Processing Time: Conventional loan approval processes involve extensive manual underwriting, often requiring 5-7 days for approval. This inefficiency hinders both lenders and borrowers. Automated machine learning models can expedite this process, evaluating applicants in real-time and significantly reducing processing time.

1.2 Importance of the Project

Improving Decision-Making Efficiency: The traditional underwriting process is often time-consuming and inconsistent. By implementing machine learning models, lenders can automate risk assessments, ensuring consistent and faster approvals while maintaining accuracy.

Enhancing Predictive Accuracy: Machine learning models outperform traditional credit scoring systems by identifying complex patterns within applicant data. This allows for a more precise risk assessment, improving overall loan approval strategies while mitigating default risks.

Scalability and Cost-Effectiveness: Automated credit assessment systems can process thousands of applications simultaneously. This scalability enhances operational efficiency, allowing financial institutions to expand their services while reducing costs associated with manual underwriting.

1.3 Real-World Applications

Commercial Banking: Banks can embed ML models into their loan origination processes, improving decision speed and accuracy.

Microfinance Institutions: ML enables financial institutions to assess creditworthiness even in the absence of traditional credit histories, fostering financial inclusion for underbanked populations.

Fintech Startups: Digital lending platforms and peer-to-peer lending services leverage machine learning for automated approvals and risk evaluations, enhancing user experience and operational efficiency.

Regulatory Compliance: Financial regulators can employ AI models to ensure lending decisions comply with ethical and legal standards, promoting transparency and accountability in financial transactions.

2 Data Overview

The dataset comprises 32,581 records with 12 features, including numerical and categorical variables. The key features are:

- **person_age:** Age of the borrower.
- **person_income:** Annual income of the borrower.
- **person_home_ownership:** Type of home ownership (e.g., rent, mortgage, own).
- **person_emp_length:** Employment length in years (with 895 missing values).
- **loan_intent:** Purpose of the loan (e.g., education, medical, etc.).
- **loan_grade:** Loan grade assigned by the lender.
- **loan_amnt:** Loan amount requested by the borrower.
- **loan_int_rate:** Interest rate on the loan (with 3,116 missing values).
- **loan_status:** Target variable indicating loan default (1) or non-default (0).
- **loan_percent_income:** Percentage of income allocated to loan payments.
- **cb_person_default_on_file:** Whether the borrower has previously defaulted on a loan.
- **cb_person_cred_hist_length:** Length of the borrower's credit history in years.

3 Exploratory Data Analysis

3.1 Loan Status Distribution

Understanding the distribution of the target variable, *loan_status*, is crucial for assessing class imbalance, which can affect model performance. Figure 1 presents the distribution of loan status in both the training and validation datasets.

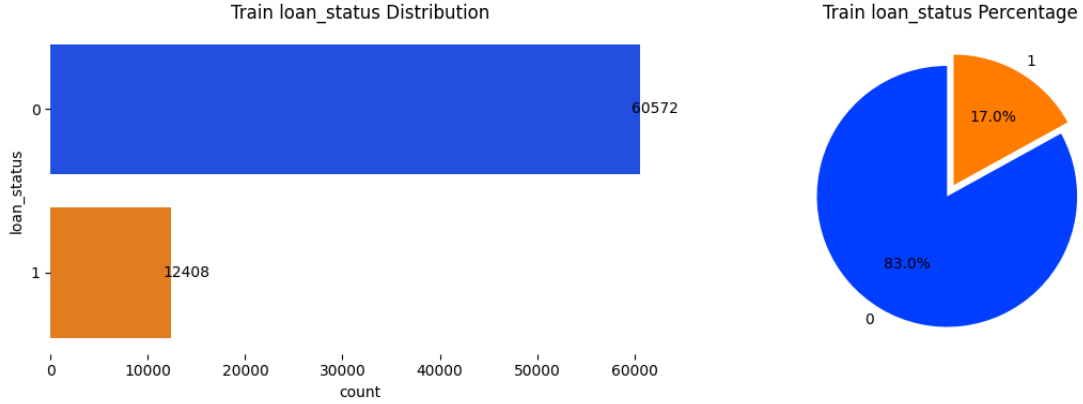


Figure 1: Distribution of Loan Status

From the distribution, we observe a significant class imbalance, with approximately 83% of the loans classified as non-default (0) and only 17% as default (1) in the training set. A similar trend is seen in the validation set, where non-default loans make up 83.3% and default loans account for 16.7%. This imbalance suggests that applying resampling techniques or adjusting class weights in model training may be necessary to improve predictive performance.

3.2 Bivariate Analysis

3.2.1 Categorical vs. Loan Status

Examining the relationship between categorical features and loan status helps identify patterns that may influence loan approval outcomes. The categorical features analyzed include:

- **Person Home Ownership:** Borrowers who rent or have mortgages show different default rates compared to those who own their homes outright.
- **Loan Intent:** Certain loan purposes, such as personal and medical loans, exhibit higher default rates than others like home improvement and education.
- **Loan Grade:** Higher-grade loans tend to have lower default rates, as expected.
- **Credit History Default:** Applicants with previous defaults on their credit history are significantly more likely to default again.

The observed trends indicate that categorical variables play a crucial role in predicting credit risk. These insights can be leveraged in feature engineering and model training to improve classification performance.

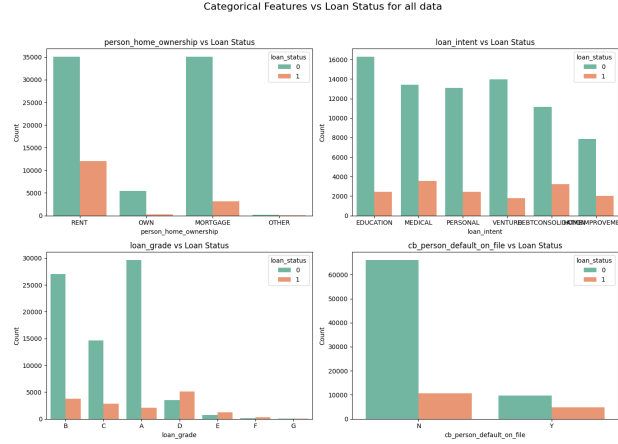


Figure 2: Comparison of Categorical features with loan status

3.2.2 Numerical vs. Loan Status

The numerical features analyzed include:

- **Person age:** Younger applicants (20–40 years) dominate the dataset, but the approval rate (loan status = 1) seems relatively consistent across age groups. Older applicants (60+ years) have fewer applications, but their approval rate appears slightly lower.
- **Person emp length:** Applicants with shorter employment lengths (0–10 years) have the highest density, but the approval rate is relatively consistent across employment lengths. Longer employment lengths (20+ years) show fewer applications, with a slight decrease in approval rates.
- **Loan int rate:** Loans with lower interest rates have a lower approval rate. As the interest rate increases, the approval rate increases significantly.
- **Cb person cred hist length:** Applicants with shorter credit histories (0–10 years) dominate the dataset, but the approval rate is relatively consistent across credit history lengths. Longer credit histories (20+ years) show fewer applications, with a slight increase in approval rates.
- **Person income:** Lower-income applicants (0–2 units) dominate the dataset, but the approval rate is relatively consistent across income levels. Higher-income applicants (4+ units) show fewer applications, with a slight increase in approval rates.
- **Loan amnt:** Smaller loan amounts (0–10,000) have the highest density and a better approval rate. As the loan amount increases (20,000+), the approval rate decreases.
- **Loan percent income:** Loans that constitute a smaller percentage of the applicant’s income (0.0–0.4) have a higher density and a better approval rate. As the loan percentage of income increases (0.6+), the approval rate decreases significantly.

Numerical Features vs Loan Status for all data (Density Plots)

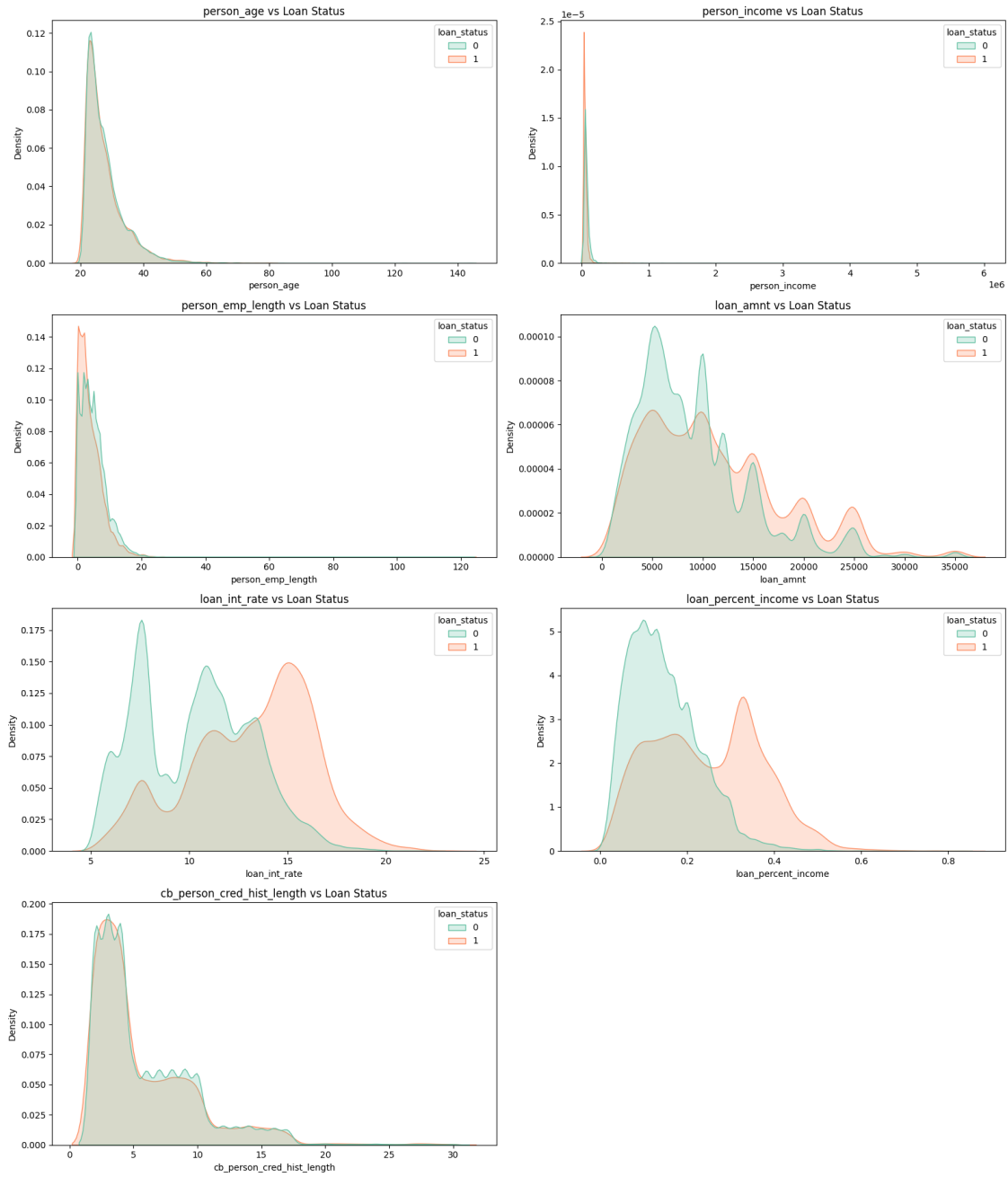


Figure 3: Comparison of Numerical features with loan status

To further explore numerical features in relation to loan status, boxplots were generated to visualize distributions, variability, and the presence of outliers.

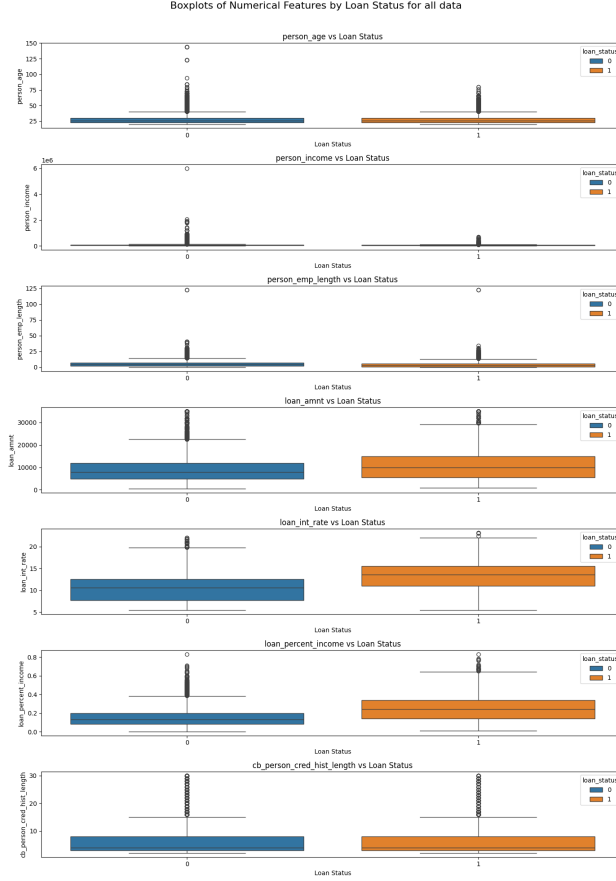


Figure 4: Boxplots of Numerical Features by Loan Status

Loan amounts and interest rates tend to be lower for approved loans, while higher loan percent income correlates with more rejections, indicating that affordability plays a crucial role in loan decisions. Outliers exist in age, income, and employment length, but they do not strongly impact approval rates. These insights aid in refining credit risk models and lending strategies.

These insights further refine model features and guide improvements in credit risk assessment.

4 Step 1: Probability of Default Modelling

Step 1 involves modeling the probability that a borrower will default on a loan. This step is crucial as it directly influences the estimation of expected losses in Step 2.

4.1 Logistic Regression

Logistic regression is a commonly used statistical model for predicting the probability of default. It estimates the probability that a borrower will default based on various independent features.

4.2 LightGBM

LightGBM (Light Gradient Boosting Machine) is an advanced boosting technique that builds decision trees efficiently. It has several advantages:

- Leaf-wise growth strategy: Splits the most promising leaf rather than growing trees level-wise.
- Histogram-based algorithm: Converts continuous features into discrete bins for faster computation.

- Gradient-based one-side sampling (GOSS): Retains high-gradient samples and randomly samples low-gradient ones.
- Exclusive feature bundling (EFB): Bundles sparse features to reduce dimensionality.

The main hyperparameters include:

- **n_estimators**: Number of trees in the ensemble.
- **learning_rate**: Controls the contribution of each tree to the final prediction.
- **scale_pos_weight**: Adjusts for class imbalance.

Performance metrics include:

- **Accuracy**: Proportion of correct predictions.
- **ROC AUC**: Measures the model's ability to distinguish between defaulters and non-defaulters.
- **F1-score**: Harmonic mean of precision and recall.

4.3 Random Forest

Random Forest is an ensemble method that constructs multiple decision trees and aggregates their outputs to improve prediction accuracy and reduce overfitting.

4.4 XGBoost

XGBoost (Extreme Gradient Boosting) is another boosting method known for its efficiency and ability to handle imbalanced datasets.

4.5 Model Comparison

The best classifier for default probability prediction was chosen based on:

- **High ROC AUC Score**: XGBoost achieved 0.9436, ensuring reliable probability estimates.
- **Balanced Precision and Recall**: XGBoost had precision = 0.9557, recall = 0.6938, and F1-score = 0.804.
- **Handling Imbalanced Data**: XGBoost provides robust performance with built-in regularization techniques.
- **Consistency Across Metrics**: LightGBM had slightly higher recall but lower precision, leading to a lower F1-score.

5 Step 2: Expected Losses Modelling

Step 2 builds on the probability of default from Step 1 to estimate the expected financial losses due to defaults. The expected loss is calculated using the formula:

$$\text{Expected Loss} = \text{Probability of Default} \times \text{Exposure at Default} \times \text{Loss Given Default} \quad (1)$$

where:

- **Probability of Default (PD)**: Estimated from Step 1.
- **Exposure at Default (EAD)**: The total value exposed to risk when default occurs.
- **Loss Given Default (LGD)**: The fraction of the exposure that will not be recovered.

5.1 Train and Test Split

The dataset is split into training and testing sets for model evaluation.

5.2 Exploratory Data Analysis (EDA) and Data Cleaning

- Fixing data types.
- Handling missing and duplicated values: Duplicates were dropped, but aggregation techniques could be explored in the future.
- Outlier treatment: Numerical transformations were applied, and all features were scaled using a Min-Max scaler.
- Box-Cox transformations were applied to targets before modeling.

5.3 Feature Engineering and Encoding

Features were encoded and transformed to optimize predictive performance.

5.4 Modeling Techniques

The following models were explored for expected loss prediction:

- **Linear Regression + Lasso**: Regularized linear regression to prevent overfitting.
- **Linear Regression + Lasso on PCA**: Principal Component Analysis (PCA) was used to reduce dimensionality before regression.
- **XGBoost**: Gradient boosting model.
- **XGBoost on PCA**: PCA was applied before XGBoost training.
- **Neural Network**: A deep learning model trained to predict expected losses.

5.5 Rationale Behind Model Selection

Different models were considered based on their strengths:

- **Linear Regression + Lasso**: Provides interpretability and is computationally efficient.
- **XGBoost**: Effective at capturing complex relationships in data.
- **Neural Network**: Can model nonlinear dependencies but requires more training data.

5.6 Model Interpretation

The final regression equation for expected losses is:

$$\begin{aligned} y = & -498.1071 + (-0.0318) \cdot \text{person_age} + (496.0818) \cdot \text{person_income} + (0.8229) \cdot \text{person_emp_length} \\ & + (-0.4599) \cdot \text{loan_grade} + (1.9194) \cdot \text{loan_int_rate} + (3.3704) \cdot \text{loan_percent_income} \\ & + (0.0278) \cdot \text{cb_person_default_on_file} + (-0.1299) \cdot \text{original} + (0.2327) \cdot \text{person_home_ownership_MORTGAGE} \\ & + (0.2215) \cdot \text{person_home_ownership_OTHER} + (0.3644) \cdot \text{person_home_ownership_OWN} \\ & + (-0.0477) \cdot \text{person_home_ownership_RENT} + (-0.1520) \cdot \text{loan_intent_DEBTCONSOLIDATION} \\ & + (0.0585) \cdot \text{loan_intent_EDUCATION} + (-0.0538) \cdot \text{loan_intent_HOMEIMPROVEMENT} \\ & + (-0.0740) \cdot \text{loan_intent_MEDICAL} + (0.0064) \cdot \text{loan_intent_PERSONAL} + (0.0420) \cdot \text{loan_intent_VENTURE} \end{aligned}$$

5.7 Final Model Comparison

Table 1 summarizes the evaluation metrics for different models.

Model	MAE	MSE	RMSE	R_squared
Lasso Regression	0.5484	0.5245	0.7242	0.4732
Lasso Regression (PCA)	0.5484	0.5245	0.7242	0.4732
XGBoost	0.0942	0.0336	0.1834	0.9662
XGBoost (PCA)	0.1619	0.0797	0.2823	0.9199
Neural Network	0.4632	0.3663	0.6053	0.6321

Table 1: Performance metrics for different models

Overall, XGBoost achieved the best performance, providing the lowest Mean Absolute Error (MAE) and highest R-squared value, indicating its superior predictive power. The application of PCA slightly reduced performance for both Lasso Regression and XGBoost. The neural network performed moderately well, but not as effectively as XGBoost.

6 Cost Feasibility

Implementing a machine learning-based loan approval system involves the following factors, each with their associated costs, such as data collection, model development and testing, infrastructure and security, and integration with banking system. Data collection and storage entails gathering the client’s data, such as applicant financial information, credit history and such and storing it securely. While specific pricing is not obtainable, cloud storage services like Amazon S3 offers prices at \$0.023 per GB for the first 50 TB per month. So scaling it to the size of a bank, and with its volume of applicants, we can project a total cost of around \$50,000.

Model development and testing entails designing, training, and validating machine learning models to predict loan approvals. We would need a few machine learning engineers to programme such models, and the average base salary for a machine learning engineer is \$7,700 a month, that comes up to \$92,000 a year. Infrastructure and Security covers expenses for cloud computing resources, data storage solutions, and implementing security measures to protect sensitive information. For instance, Microsoft Azure’s general-purpose instances start at \$58.40 per month for an A1 instance with 1.75 GB RAM and 225 GB of temporary storage. This would add up to approximately \$75,000 annually.

Integration with banking systems involves developing APIs and ensuring seamless integration with existing banking platforms and third-party services. Specific costs for API development and system integration can vary widely based on the complexity and specific requirements of the banking systems involved. While exact figures are not always publicly available, we can estimate that it would cost around \$50,000 annually. As such, the total estimated Cost ranges between \$275,000 and \$300,000, and implantation would bring about 30% increase in approval efficiency and up to 20% reduction in loan defaults.

6.1 Conclusion

Our report presented a two-step modeling approach for estimating **Probability of Default (PD)** and **Expected Losses (EL)** in loan approvals.

6.1.1 Probability of Default Modeling

- Several machine learning models were evaluated, including **Logistic Regression, LightGBM, Random Forest, and XGBoost**.
- **XGBoost** was identified as the best-performing model, achieving the highest **ROC AUC score (0.9436)** and a well-balanced trade-off between **precision and recall**, making it suitable for default risk prediction.

6.1.2 Expected Losses Modeling

- Various models were tested, including **Lasso Regression**, **XGBoost**, and **Neural Networks**, with and without **Principal Component Analysis (PCA)**.
- **XGBoost** again demonstrated superior performance with the lowest **Mean Absolute Error (MAE)** = **0.0942** and highest **R-squared** = **0.9662**, making it the most reliable choice for predicting expected financial losses.

6.1.3 Key Takeaways

- **XGBoost consistently outperformed other models** in both PD and EL modeling, showing its robustness in handling structured financial datasets.
- **Dimensionality reduction with PCA** had mixed effects—while it reduced computation time, it also slightly impacted predictive accuracy.
- **Neural Networks** performed reasonably well but did not surpass XGBoost, possibly due to limited training data or hyperparameter tuning constraints.

6.1.4 Future Improvements

- **Feature Engineering Enhancements:** Introducing additional financial indicators or alternative embeddings could improve model interpretability.
- **Hybrid Models:** Combining statistical methods with deep learning may provide further accuracy gains.
- **Real-Time Adaptation:** Implementing an adaptive model that continuously learns from new loan data could enhance predictive power in dynamic financial markets.

Overall, our study highlights the effectiveness of **XGBoost** in loan risk assessment and expected loss modeling, offering a practical tool for financial institutions to **mitigate default risks and optimize loan approval strategies**.

7 Reference

1. Kaggle Playground Series - S4E10
2. Azure Cloud Services Pricing
3. Glassdoor - Machine Learning Engineer Salaries